

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech. Examination (CE/IT)

December 2013

CE301 Parallel Computing

Date: 20.11.2013, Wednesday

Time: 10:00 am To 01:00 pm

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Do not write content of Header.h file for any program.

SECTION-I

Q-1 Answer the following questions. [11]

- [A] What do you mean by quasi dynamic temporal parallelism? [2]
- [B] What are the arguments and functionality of user function call: `Process_fork()`? [2]
- [C] Write the differences between Process and Thread. [2]
- [D] Why is thread called light weight process? [2]
- [E] What are the applications of Parallel Computing? [3]

Q-2 Answer the following questions. [12]

- [A] Write down the difference between data parallelism and temporal parallelism. [5]
- [B] Write a 'C' program to create histogram from an array using loop-splitting. [7]

OR

- [B] Write and explain barrier function call. [7]

Q-3 Answer the following questions.(Attempt any Two) [12]

- [A] Write a 'C' program to sum a two array of N elements using self-scheduling.
- [B] Write a 'C' program using thread to illustrate Producer-Consumer Problem
- [C] Write programs which explain working of Expression Splitting.

SECTION-II

Q-4 Answer the following questions. [11]

- [A] What do you mean by joinable thread? [2]
- [B] What are the factors affects the ideal speedup? [2]
- [C] What are the attributes of thread? [2]
- [D] Explain how joining of threads differs from joining of processes. [2]
- [E] List the requirement of the characteristics of operating system to support distributed computing? [3]

Q-5 Answer the following questions. [12]

- [A] Write a thread program to sum an array of N elements using loop splitting. [5]
- [B] Which are the system call available to make communication among multiple threads? [7]

OR

- [B] What is RPC? How does it communicate with other nodes in network? [7]

Q-6 Answer the following questions.(Attempt any Two) [12]

- [A] Draw architecture of DCE. Explain services offered by DCE.
- [B] Explain the architecture of PVM and explain terminology used in PVM.
- [C] Explain in detail how to create thread and join thread.